Prasad Reddy

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PROFESSIONAL SUMMARY

Results-driven Data Engineer with 3+ years of experience designing and delivering scalable cloud-native data pipelines, RESTful APIs, and big data platforms across the automotive and financial services industries. Deep expertise in GCP (BigQuery, Dataproc, Cloud Run, Cloud Functions) and Python-based backend development. Proven ability to architect high-throughput ETL workflows, integrate enterprise data systems, and optimize production workloads in Agile delivery environments.

SKILLSETS

- **GCP Services**: BigQuery, Cloud Run, Dataproc, Cloud Functions, Cloud Storage, Pub/Sub, Cloud Composer, Cloud Scheduler, Vertex AI, Secret Manager
- **Languages**: Python, SQL, PySpark, PL/SQL, Bash
- **Frameworks / API**: FastAPI, Flask, SQLAlchemy ORM, Swagger / OpenAPI, RESTful API design
- **Big Data**: Apache Spark, Apache Airflow, Databricks, Kafka, Hadoop, Hive
- **Cloud Platforms**: Google Cloud Platform (GCP)
- **Databases**: BigQuery, PostgreSQL, Oracle, SQL Server, Snowflake, Cosmos DB, DynamoDB
- **DevOps / IaC**: Terraform, Docker, Git, GitLab CI/CD, Prefect, CI/CD pipelines
- **Python Libraries**: Pandas, NumPy
- **Methodology**: Agile / Scrum, Jira, Confluence, Code Reviews, PEP8, TDD

WORK EXPERIENCE

Ford Motor Company — **Apr 2024 - Present**

Python Data Engineer — *Cleveland, OH (Remote)*

Ford Motor Company is a global automotive leader manufacturing vehicles and mobility solutions across 190+ markets. The project involved modernizing Ford's enterprise data infrastructure by migrating a legacy flat-file-based data system to a scalable hub-and-spoke architecture on GCP, enabling real-time ingestion, centralized data governance, and self-service analytics for manufacturing and connected vehicle teams.

- Led the migration from a legacy flat-file system to a hub-and-spoke architecture on GCP centralized raw data ingestion into Cloud Storage (hub), with spoke pipelines feeding domain-specific BigQuery datasets for manufacturing, supply chain, and connected vehicle analytics teams.
- Designed and implemented Cloud Functions to automate file-arrival triggers on Cloud Storage each new flat-file drop (CSV/JSON) automatically invokes a serverless function to validate schema, apply transformations using Pandas, and route clean records to the appropriate BigQuery spoke table.
- Built additional Cloud Functions for lightweight event-driven tasks: dead-letter queue handling for failed ingestion events, Pub/Sub message fan-out to downstream consumers, and automated metadata tagging of ingested datasets in Data Catalog.
- Developed PySpark transformation jobs on Dataproc for heavy-volume batch processing ingest raw vehicle telemetry and manufacturing data.
- Applied cleansing and normalization using Pandas for in-memory profiling and validation, then write output to partitioned Parquet in Cloud Storage before loading to BigQuery.
- Architected BigQuery spoke datasets with partitioned and clustered tables per domain reduced query costs by ~35% and improved analytical query performance for downstream Looker Studio dashboards tracking production KPIs.
- Implemented Apache Airflow DAGs (Cloud Composer) to orchestrate the full hub-and-spoke pipeline dependency chains, retry logic, SLA alerting, and conditional branching based on file type and source system.
- Provisioned and managed GCP infrastructure using Terraform - Cloud Run services, Dataproc clusters, Cloud Functions, IAM bindings, and Cloud Storage buckets, enforced GitLab CI/CD gates with Bandit, detect-secrets, and tfsec security scans.
- Authored pytest test suites covering Cloud Function logic, Pandas transformation steps, achieved 80%+ unit test coverage; integrated tests into CI/CD pipeline to block deployments on regression failures.

PNC — **Jul 2022 - Dec 2023**

Python Developer — *Hyderabad, India*

Designed and developed scalable RESTful APIs, enabling seamless integration with front-end applications and external services.

- Built and maintained database models using SQLAlchemy ORM, ensuring efficient data structure, integrity, and performance.
- Defined complex relationships between models, supporting business logic and optimizing database queries.
- Integrated with Oracle databases, writing efficient SQL queries and managing schema migrations to support application evolution.
- Generated interactive API documentation via Swagger UI, improving developer experience and API usability.
- Implemented input validation, error handling, and response formatting to ensure secure and consistent API behavior.
- Collaborated with cross-functional teams (e.g., frontend, QA, DevOps) to align development goals with product requirements.
- Wrote unit and integration tests for APIs and data layers to ensure reliability and support CI/CD pipelines.
- Optimized API performance by profiling endpoints, caching strategies, and tuning SQL queries.
- Maintained code quality and consistency through code reviews, version control and adherence to PEP8 standards.
- Provided L2/L3 production support, promptly diagnosing and resolving issues related to API failures, database connectivity, and performance degradation.

EDUCATION DETAILS

Jawaharlal Nehru University
Bachelor of Engineering, Electronics and Communication Engineering

Wilmington University
Master's, information systems technology